INCH-POUND

MIL-C-15370/4C
22 September 1989
SUPERSEDING
MIL-C-15370/4B
5 June 1975

MILITARY SPECIFICATION SHEET

COUPLERS, DIRECTIONAL (WAVEGUIDE, MULTIHOLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-15370.

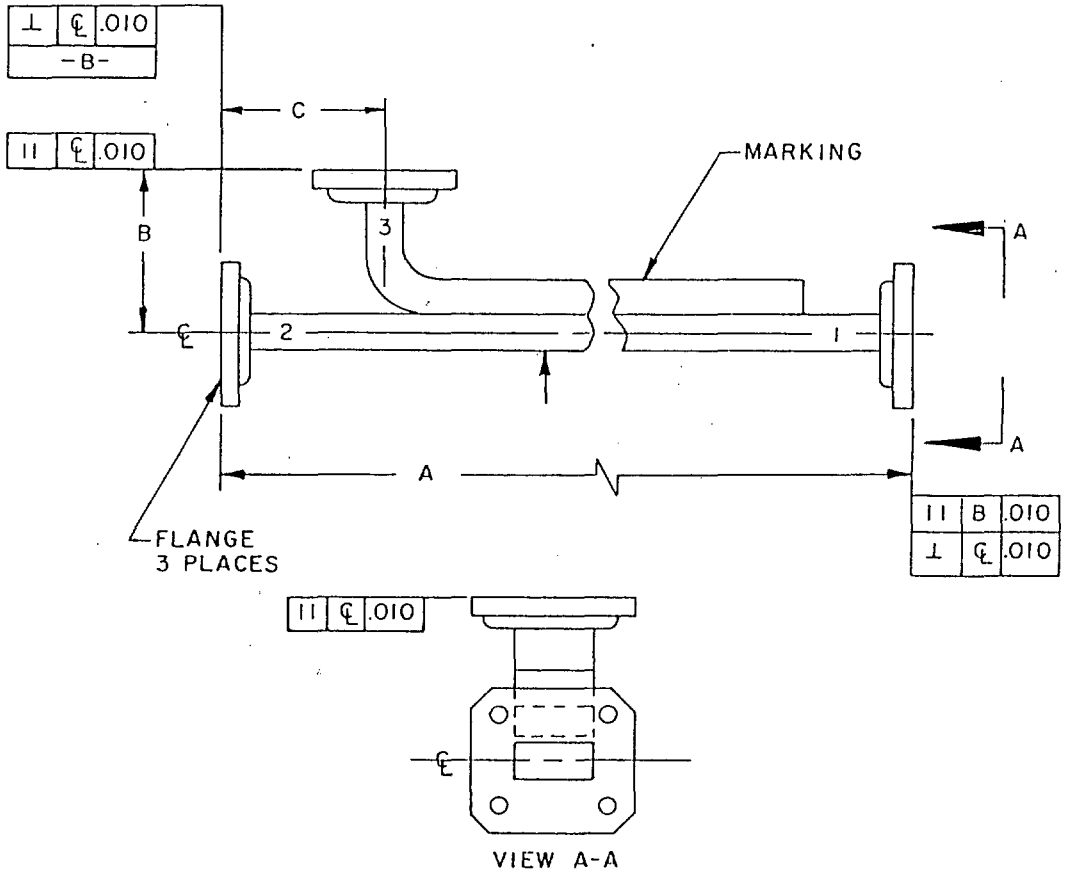


FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design, construction and physical dimensions: See figure 1 and table I.

Equivalent waveguide: See table I.

Flange: See table I.

Operating temperature range: -40°C to +85°C for dash numbers -001 through -017.
-20°C to +65°C for dash numbers -018 through -027.

Frequency range: See table II.

Coupling: See table II.

Coupling tolerance: See table II.

Coupling variation: See table II.

Effective directivity: See table II.

VSWR: See table II.

Marking: See figure 1.

Cross-reference data: See table III.

Part number: M15370/4- (and a dash number from table I).

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TABLE I. Physical characteristics.

Dash number	Dimensions 1/ 2/ 3/						Equivalent waveguide	Flange
	A		B		C			
	Min	Max	Min	Max	Min	Max		
001N S	26.437 (671.50)	26.563 (674.70)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)	M85/1-067 (RG-50/U)	M3922/55-001 (UG-344/U)
002N S	25.500 (647.70)	26.625 (676.28)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)	M85/1-067 (RG-50/U)	M3922/55-001 (UG-344/U)
003N S	25.500 (647.70)	26.625 (676.28)	4.500 (114.30)	4.625 (118.48)	2.000 (50.80)	2.125 (53.98)	M85/1-067 (RG-50/U)	M3922/55-001 (UG-344/U)
004N S	18.563 (471.50)	19.188 (487.38)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
005N S	17.438 (442.90)	17.563 (446.10)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
006N S	17.438 (442.90)	17.563 (446.10)	2.938 (74.63)	3.063 (77.80)	1.063 (27.00)	1.188 (30.18)	M85/1-073 (RG-51/U)	M3922/53-002 (UG-51/U)
007N S	16.625 (422.28)	16.750 (425.45)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)
008N S	15.625 (396.88)	15.750 (400.05)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)	M85/1-079 (RG-52/U)	M3922/55-001 (UG-39/U)
009N S	15.625 (396.88)	15.750 (400.05)	2.938 (74.63)	3.063 (77.80)	1.000 (25.40)	1.125 (28.58)	M85/1-079 (RG-52/U)	M3922/55-001 (UG-39/U)
010N S	12.188 (309.58)	12.313 (312.75)	1.968 (50.00)	2.094 (52.79)	0.687 (17.48)	0.813 (20.68)	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)
011N S	9.313 (236.55)	10.656 (270.66)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
012N S	9.313 (236.55)	9.968 (253.19)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
013N S	7.562 (192.10)	9.968 (253.19)	1.937 (49.20)	3.000 (76.20)	0.562 (14.27)	1.000 (25.40)	M85/1-106 (RG-66/U)	M3922/54-001 (UG-595/U)
014N S	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)	M85/1-109 (RG-349/U)	M3922/53-006 (UG-1665/U) 4/
015N S	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)	M85/1-090 (RG-349/U)	M3922/53-012
016N S	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)	M85/1-090 (RG-349/U)	M3922/53-012
017N S	20.350 (516.89)	20.410 (518.41)	2.230 (56.64)	2.270 (57.66)	0.980 (24.89)	1.020 (25.91)	M85/1-090 (RG-349/U)	M3922/53-012
018N S	5.975 (151.77)	8.750 (222.25)	1.630 (41.40)	1.938 (49.23)	0.400 (10.16)	0.719 (18.26)	M85/3-008 (RG-271/U)	M3922/54-003 (UG-599/U)

See footnotes at end of table.

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TABLE I. Physical characteristics - Continued.

Dash number	Dimensions 1/ 2/ 3/						Equivalent waveguide	Flange
	A		B		C			
	Min	Max	Min	Max	Min	Max		
019N S	5.975 (151.77)	8.750 (222.25)	1.630 (41.40)	1.938 (49.23)	0.400 (10.16)	0.719 (18.26)	M85/3-006 (RG-96/U)	M3922/54-003 (UG-599/U)
020N S	7.20 (182.88)	7.40 (187.96)	1.51 (38.35)	1.55 (39.37)	0.782 (19.86)	0.820 (20.83)	M85/1-078	M3922/53-003 (UG-135/U) 5/
021N S	7.20 (182.88)	7.40 (187.96)	1.51 (38.35)	1.55 (39.37)	0.782 (19.86)	0.820 (20.83)	M85/1-077 (RG-67/U)	M3922/59-008 (UG-136B/U) 6/
022N S	7.20 (182.88)	7.40 (187.96)	1.51 (38.35)	1.55 (39.37)	0.782 (19.86)	0.820 (20.83)	M85/1-077 (RG-67/U)	7/
023N S	13.50 (342.90)	13.95 (354.15)	1.875 (47.63)	1.879 (47.73)	0.688 (17.48)	0.706 (17.93)	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)
024N S	12.40 (314.96)	12.45 (316.23)	1.875 (47.63)	1.879 (47.73)	0.688 (17.48)	0.706 (17.93)	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)
025N S	12.40 (314.96)	12.45 (316.23)	1.875 (47.63)	1.879 (47.73)	0.688 (17.48)	0.706 (17.93)	M85/1-093 (RG-107/U)	M3922/53-005 (UG-419/U)
026N S	12.40 (314.96)	13.01 (330.45)	1.370 (34.80)	1.380 (35.05)	0.688 (17.48)	0.706 (17.93)	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)
027N S	12.40 (314.96)	15.00 (381.00)	1.630 (41.40)	1.938 (49.23)	0.400 (10.16)	0.719 (18.26)	M85/1-089 (RG-91/U)	M3922/53-005 (UG-419/U)

1/ Dimensions are in inches.

2/ Metric equivalents are given for general information only.

3/ Metric equivalents are in parentheses.

4/ Except the lip thickness is .25 inch and tapped holes .138-32UNC-2B.

5/ Except for tapped holes .164-32UNC-2B in the input (1) flange.

6/ Except for clearance holes of .167/179 dia in the output (2) flange.

7/ Input flange (1) is M3922/53-003, and the output and coupled flanges (2 and 3) are M3922/59-008. All three flanges have threaded holes .164-32UNC-2B.

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TABLE II. Electrical characteristics.

Dash number	Frequency range	Coupling (dB)		Coupling variation (dB) max	Effective directivity (dB) min	VSWR max		Power dissipation of internal load (avg)
		Nominal	Tolerance			Primary line	Secondary line	
	GHz							Watts
001N S	5.85-8.2	3	±0.5	±0.5	40	1.10:1	1.20:1	1
002N S	5.85-8.2	10	±0.5	±0.5	40	1.05:1	1.15:1	1
003N S	5.85-8.2	20	±0.5	±0.5	40	1.05:1	1.15:1	1
004N S	7.05-10.0	3	±0.5	±0.5	40	1.10:1	1.20:1	1
005N S	7.05-10.0	10	±0.5	±0.5	40	1.05:1	1.15:1	1
006N S	7.05-10.0	20	±0.5	±0.5	40	1.05:1	1.15:1	1
007N S	8.2-12.4	3	±0.5	±0.5	40	1.10:1	1.20:1	1
008N S	8.2-12.4	10	±0.5	±0.5	40	1.05:1	1.15:1	1
009N S	8.2-12.4	20	±0.5	±0.5	40	1.05:1	1.15:1	1
010N S	12.4-18.0	20	±0.5	±0.5	40	1.05:1	1.15:1	1
011N S	18.0-26.5	3	±0.5	±0.5	40	1.10:1	1.20:1	0.5
012N S	18.0-26.5	10	±0.5	±0.5	40	1.05:1	1.15:1	0.5
013N S	18.0-26.5	20	±0.5	±0.5	40	1.05:1	1.15:1	0.5
014N S	11.0-17.0	10	±0.4	±0.4	35	1.10:1	1.20:1	1
015N S	11.0-17.0	15	±0.4	±0.4	35	1.10:1	1.20:1	1
016N S	11.0-17.0	20	±0.4	±0.4	35	1.10:1	1.20:1	1
017N S	11.0-17.0	30	±0.4	±0.4	35	1.10:1	1.20:1	1

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TABLE II. Electrical characteristics - Continued.

Dash number	Frequency range	Coupling (dB)		Coupling variation (dB) max	Effective directivity (dB) min	VSWR max		Power dissipation of internal load (avg)
		Nominal	Tolerance			Primary line	Secondary line	
	GHz							
018NS	26.5-40.0	20	±0.7	±0.6	40	1.05:1	1.20:1	0.5
019NS	26.5-40.0	30	±0.7	±0.6	40	1.05:1	1.20:1	0.5
020NS	8.2-12.4	20	±0.5	±0.5	30	1.10:1	1.20:1	3
021NS	8.2-12.4	20	±0.5	±0.5	40	1.05:1	1.15:1	1
022NS	8.2-12.4	40	±0.4	±0.4	30	1.05:1	1.15:1	1
023NS	12.4-18.0	3	±0.5	±0.5	40	1.10:1	1.20:1	1
024NS	12.4-18.0	6	±0.5	±0.5	40	1.10:1	1.20:1	1
025NS	12.4-18.0	10	±0.5	±0.5	40	1.10:1	1.20:1	1
026NS	12.4-18.0	10	±0.5	±0.5	30	1.10:1	1.20:1	1
027NS	12.4-18.0	40	±0.5	±0.5	40	1.10:1	1.20:1	1

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TABLE III. Cross-reference of part number/AN nomenclature.

Dash number	AN nomenclature
001	CU-1510/U
002	CU-1511/U
003	CU-1512/U
004	CU-1507/U
005	CU-1508/U
006	CU-1509/U
007	CU-1513/U
008	CU-1514/U
009	CU-1515/U
010	CU-1519/U
011	CU-1522/U
012	CU-1523/U
013	CU-1524/U

NOTE: Directional couplers covered by this specification sheet are substitutable for the manufacturer's part numbers specified in table IV. This information in no way implies that the manufacturer's part number or NSN is suitable for the military part number.

TABLE IV. Substitution data.

Military part number M15370/4-	Manufacturer's code	Manufacturer's part number	NSN 5985-
001	16055	C201A	00-679-0637
001	28480	J752A	00-679-0637
002	28480	J752C	00-679-0638
003	16055	C201D	00-679-0639
003	28480	J752D	00-679-0639
004	28480	H752A	00-679-0630
005	28480	H752C	00-729-6971
006	28480	H752D	00-814-4721
007	28480	X752A	00-838-9920
007	00341	DD174X1-3DB	00-838-9920
008	94117	1292022	00-646-9472
008	28480	X752C	00-646-9472
008	07397	20-682129-1	00-220-9728
008	01456	90CT106-1E	00-220-9728
008	82577	927105-1B	01-156-8086
008	01456	90CT336-1-C	01-156-8086
008	00443	673-10	01-141-4651
008	00929	X610C	00-364-3668
009	19200	7904840	00-679-0645
009	80006	414796	00-679-0645
009	28480	X752D	00-679-0645
009	00929	X610D	00-840-2124
010	28480	P752D	00-562-8870
010	89372	ST4130201HE0002	00-467-1460
010	00443	774-20	00-467-1460

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TABLE IV. Substitution data - Continued.

Military part number M15370/4-	Manufacturer's code	Manufacturer's part number	NSN 5985-
011	00929	K610A	
011	28480	K752A	00-859-7057
011	03000	5985-16150	00-859-7057
012	00929	K610C	00-692-8714
012	28480	K752C	00-913-9842
013	00929	K610D	
013	28480	K752D	00-084-8992
014	05869	712402-1	
014	00443	70467P1	
015	05869	712402-2	
015	00443	70467P2	
016	05869	712402-3	
016	00443	70467P3	
017	05869	712402-4	
017	00443	70467P4	
018	28480	R752D	00-084-8991
018	94990	58-P06571-D	
019	34228	885C052581	01-037-3349
019	96341	MA-772C	
020	19315	2132053-1	
020	01456	90CT579	
021	01456	90LT79	00-007-3677
021	49956	480104-1	00-007-3677
022	82918	380-10260-1	
022	01456	90CT136-20P-4E	
023	00929	Y610A	00-604-3327
023	28480	P752A	00-604-3327
024	00443	774-6	00-503-9130
024	89372	ST4130201HE0001	00-503-9130
025	00929	Y610C	00-887-9933
026	06424	10205-1	00-213-6366
026	00443	774-10	00-213-6366
026	28480	P752	00-226-6846
026	99899	1069-10	00-226-6846
026	80058	CU673U	00-226-6846
027	96214	806330-1	01-149-8952
027	00443	774-40	01-149-8952

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NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Navy - CG, MC, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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